OBJECT ORIENTED SOFTWARE ENGINEERING

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Smoke Testing

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Smoke Testing

- Smoke testing is one of the popular high level test performed on **initial build** that undergo various software tests.

- It is a software testing services performed **after software build** to **find if the critical functionalities of the programs are working fine.**

Smoke Testing

- Build is a version of a software application which is still under testing.
- It is a piece of software application which is accountable for some **particular functionalities** in a software product

Smoke Testing

- The smoke testing is executed before any **detailed functional** or **regression tests** are conducted on that particular software build (or code).
- In this testing method, the **development team deploys the build in QA**. The subsets of test cases are taken, and then testers run test cases on the build.
- The **QA team test the application against the critical functionalities**. These series of **test cases are designed to expose errors** that are in build. If these tests are passed, QA team continues with **Functional Testing.**

Smoke Testing

- Any failure indicates a need to **handover the system back to the development team**.
- **Whenever there is a change in the build, we perform Smoke Testing to ensure the stability.**
- **Example**: -New registration button is added in the login window and build is deployed with the new code. We perform smoke testing on a new build.

Smoke Testing

- **Smoke testing** also known as "**Build Verification Testing**" - is a type of software testing that comprises of a non-exhaustive set of tests that aim at ensuring that the most important functions work.

- The results of this testing is used to decide **if a build is stable enough to proceed with further testing**
  - (A build includes all **data files, libraries, reusable module & engineered component** that are required to implement one or more product function).

Smoke Testing

- Smoke Testing is a testing technique that is **inspired from hardware testing**, which **checks for the smoke** from the hardware components once the hardware's power is switched on. Similarly in Software testing context, smoke testing refers **to testing the basic functionality of the build**.

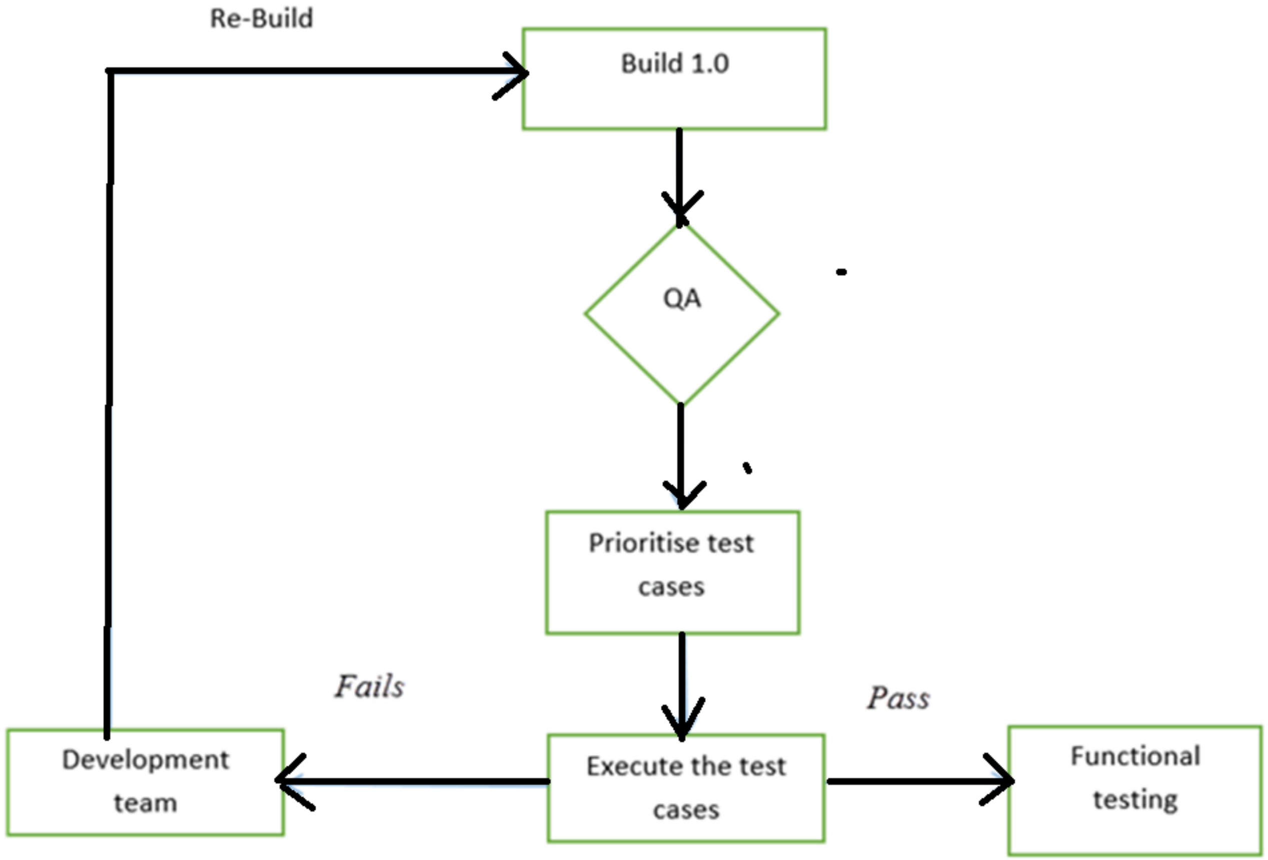
- If the **Test fails**, build is declared as **unstable** and it is NOT tested anymore until the smoke test of the build passes.

Smoke Testing Features

- Identifying the **business critical functionalities** that a product must satisfy.
- **Designing and executing the basic functionalities** of the application.
- Ensuring that the **smoke test passes each and every build** in order to proceed with the testing.
- Smoke Tests **enables uncovering obvious errors** which **saves time and effort** of test team.
- Smoke Tests can be **manual or automated.**

Smoke Test Cycle

- Below flow chart shows how Smoke Testing is executed. Once the build is deployed in QA and, smoke tests are passed we proceed for functional testing. If the smoke test fails, we exit testing until the issue in the build is fixed.



Smoke Testing…

- The result of smoke test is used to decide **whether to proceed with further testing.**
  - If the smoke test passes, go ahead with further testing. If it fails, **halt further tests** and ask for a new build with the required fixes.
- If an application is badly broken, detailed testing might be a waste of time and effort.

Smoke Testing…

Advantages:

- Easy to perform testing
- Defects will be identified in early stages.
- Improves the quality of the system
- Reduces the risk
- Progress is easier to access.
- Saves test effort and time
- Easy to detect critical errors and correction of errors.
- It runs quickly
- Minimizes integration risks

Smoke Testing…

Levels applicable to:

- Smoke testing is normally used in
  - **Integration Testing,**
  - **System Testing and**
  - **Acceptance Testing levels**

Smoke Testing Vs Regression Testing

Regression testing	Smoke testing
Regression testing is used to check defects generated to other modules by making the changes in existing programs.	At the time of developing a software product smoke testing is used.
In regression tested components are tested again to verify the errors.	It permit the software development team to test projects on a regular basis.
Regression testing needs extra manpower because the cost of the project increases.	Smoke testing does not need an extra manpower because it does not affect the cost of project.
Testers conduct the regression testing.	Developer conducts smoke testing just before releasing the product.

THANK YOU

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